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| **Fiscal Year:** | FY 2026 | **Task Last Updated:** | FY 11/15/2025 |
| **PI Name:** | Diaz Artiles, Ana Ph.D. | | |
| **Project Title:** | Dose-Response Curves of Cardiovascular and Ocular Variables During Graded Lower Body Negative Pressure (LBNP) in Microgravity Conditions Using Parabolic Flight | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | (1) **HHC**:Human Health Countermeasures | | |
| **Human Research Program Risks:** | (1) **Cardiovascular**:Risk of Cardiovascular Adaptations Contributing to Adverse Mission Performance and Health Outcomes<br>(2) **SANS**:Risk of Spaceflight Associated Neuro-ocular Syndrome (SANS) | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Zip Code:** | 77843-0001 | **Congressional District:** | 17 |
| **Comments:** | | | |
| **Project Type:** | Flight,Ground | **Solicitation / Funding Source:** | 2024 HERO 80JSC024NA001-FLAGSHIP, OMNIBUS, and NSCOR Crew Health: Topics in Human Health Countermeasures, Behavioral Performance, and Space Radiation and NASA Human Research Program Omnibus Opportunity |
| **Start Date:** | 01/01/2025 | **End Date:** | 12/31/2026 |
| **No. of Post Docs:** | 1 | **No. of PhD Degrees:** | |
| **No. of PhD Candidates:** | 6 | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | | **Monitoring Center:** | NASA JSC |
| **Contact Monitor:** | Brocato, Becky | **Contact Phone:** | |
| **Contact Email:** | becky.brocato@nasa.gov | | |
| **Flight Program:** | Parabolic | | |
| **Flight Assignment:** | NOTE: End date changed to 12/31/2026 per C. Ribeiro/HHC (Ed., 4/16/26) | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Whittle, Richard Ph.D. ( University of California, Davis ) | | |
| **Grant/Contract No.:** | 80NSSC25K7363 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |
| **Task Description:** | Exposure to weightlessness results in the removal of hydrostatic pressure gradients and a headward fluid shift. This has currently unknown consequences, but it has been hypothesized to contribute to a series of neuro-ocular structural and functional changes occurring during spaceflight, collectively known as Spaceflight Associated Neuro-ocular Syndrome (SANS). Additionally, recently demonstrated stagnant and retrograde blood flow and venous thrombosis in the left internal jugular vein (IJV) during spaceflight could also be associated with sustained headward fluid shift and removal of hydrostatic gradients. Thus, countermeasures focused on reversing the headward fluid shift, such as lower body negative pressure (LBNP), become particularly relevant. Previous studies on the ground (i.e., Earth conditions, mostly in supine position or 6° head-down tilt) investigating the effects of LBNP have shown promising results. However, there are fundamental differences in physiological responses between Earth and microgravity conditions (e.g., presence/absence of front-to-back, or Gx, acceleration). Thus, ground results might not be directly extrapolated to true microgravity conditions.<br>The objective of this analog-based research effort is to generate acute dose-response curves of a large set of cardiovascular, autonomic, cephalad, and ocular measures as a function of LBNP level, during true microgravity conditions, using parabolic flight. Participants (in supine position with respect to the airplane framework) will be exposed to multiple levels of LBNP during microgravity exposure. At each LBNP level, we will collect a large variety of metrics to characterize and quantify the fluid shift induced by LBNP. These measures include hemodynamic and autonomic responses, cerebral blood flow, leg volume at the calf level, lower body water content, intraocular pressure (IOP), and IJV and common carotid artery (CCA) characteristics. In addition, these responses will be compared to similar responses collected on the ground, where the Gx acceleration is present. Importantly, our study will generate the most comprehensive and systematic dataset ever collected on LBNP responses in true microgravity conditions, providing unique insight into the role of gravity in the Gx body axis. We expect that our study will contribute to the development of LBNP as an operational countermeasure for both SANS and spaceflight-associated venous thromboembolism events. | | |
| **Rationale for HRP Directed Research:** | | | |

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| **Research Impact/Earth Benefits:** | Our study will generate a more inclusive conceptual framework for fluid-shift response prediction to LBNP interventions on Earth and in microgravity. This framework will improve our understanding of LBNP as a potential operational countermeasure, providing a unique insight into the effects of tissue compressive forces in the Gx body axis during LBNP interventions on Earth. Our results will also contribute to the assessment of the pathogenesis of SANS and spaceflight venous thromboembolism events. |
| **Task Progress:** | This project aims to characterize cardiovascular, autonomic, and ocular responses to graded Lower Body Negative Pressure (LBNP) during true microgravity conditions achieved through parabolic flight. Microgravity causes headward fluid shifts that can alter cardiovascular function and may contribute to ocular changes and other physiological adaptations. LBNP is a promising countermeasure designed to mitigate these effects by generating hydrostatic gradients in the lower body, potentially reducing headward fluid shifts, cardiovascular deconditioning, and risks associated with spaceflight-associated neuro-ocular syndrome (SANS) and venous thromboembolism events.<br>During this reporting period, we successfully completed all preparatory milestones, including ethical and regulatory approvals in both the United States and France, as well as integration and flight certification of the LBNP system. All instrumentation required for the study, including cardiovascular monitors, cerebral and calf blood flow sensors, ultrasound devices, ocular measurement tools, and bioimpedance systems, was fully tested and verified for electromagnetic compatibility and flight safety.<br><br>The first parabolic flight campaign (87th ESA campaign) was conducted in September 2025 aboard the Airbus A310 Zero-G aircraft operated by Novespace in Mérignac, Bordeaux, France. This campaign marked the first in-flight implementation of the study protocol, with participants exposed to graded LBNP levels while supine during repeated parabolas. Data collection included comprehensive cardiovascular, autonomic, cerebral, ocular, and peripheral fluid measurements under both 1g and 0g conditions. Baseline measurements and pre-flight participant assessments were completed to ensure robust paired datasets.<br><br>A first set of participants was successfully tested during this campaign, with most planned physiological measurements collected as intended. The integrated instrumentation suite performed reliably, and the data obtained provide a strong foundation for analysis and future parabolic flight campaigns. Since completion of the flight, we have focused on data management, including cleaning and organizing the multi-instrument datasets, and data analysis is ongoing to process and align measurements across instruments, LBNP levels, and gravity phases.<br><br>We are also preparing for participation in upcoming ESA parabolic flight campaigns. These future flights will build on the experience gained during the first campaign, allowing us to expand the dataset, refine analysis pipelines, and improve the robustness of our findings.<br><br>Overall, the first campaign represents a major milestone for the project. It demonstrated the feasibility of the experimental protocol in true microgravity conditions, confirmed the successful integration of multiple measurement systems, and provided the foundation for collecting complex physiological datasets that will inform future countermeasure development for spaceflight.<br><br>These results directly support NASA's Human Research Program by advancing our understanding of cardiovascular, autonomic, and ocular responses to microgravity and evaluating LBNP as a potential countermeasure. The data collected during this first campaign provide critical insight into the effects of hydrostatic gradients on headward fluid shifts, cardiovascular deconditioning, and risks associated with SANS and venous thromboembolism. Establishing reliable measurement protocols and demonstrating the feasibility of collecting complex physiological datasets in true microgravity lays the foundation for future operational studies, hardware development, and evidence-based countermeasure strategies to support astronaut health on long-duration space missions. |
| **Bibliography Type:** | Description: (Last Updated: 07/10/2026) |
| **Significant Media Coverage** | La Vanguardia (featuring Dr. Ana Diaz Artiles and Huc Pentinat Llurba). "Mujeres científicas investigan para proteger la salud cardiovascular y ocular de los astronautas." La Vanguardia, September 15 2025. https://www.lavanguardia.com/sociedad/20250915/11060332/mujeres-cientificas-investigan-proteger-salud-cardiovascular-ocular-astronautas-agenciaslv20250915.html , Sep-2025 |
| **Significant Media Coverage** | Objetivo Planeta (featuring Dr. Ana Diaz Artiles and Huc Pentinat Llurba). "Vuelos parabólicos: ingravidez en la Tierra." Objetivo Planeta. National TV channel TVE. November 11 2025. https://www.rtve.es/play/videos/objetivo-planeta/vuelos-parabolicos-ingravidez-tierra/16806609/ , Nov-2025 |
| **Significant Media Coverage** | Schaechinger A. (Texas A&M University). "The ups and downs of space research." Texas A&M University ( https://engineering.tamu.edu/news/2025/11/the-ups-and-downs-of-space-research.html) , Nov-2025 |